CHM 101

ELECTROCHEMISTRY
3rd Lecture

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Galvanic cells

The simple cell is also called a voltaic cell in honour of Allessandro Volta who in 1800 invented the voltaic pile that was regarded as the first battery. Even though a battery consists of multiple cells, the words battery and a simple Galvanic cell are commonly used interchangeably.

The simple cell is not rechargeable, and it consists of two metals dipped into an electrolyte. The Daniell cell is an example of a simple or Galvanic cell. It consists of a zinc rod dipped into zinc sulphate solution as the anode and a copper rod dipped into copper sulphate solution as the cathode. A salt bridge is incorporated to mop up excess ions at the two electrodes.

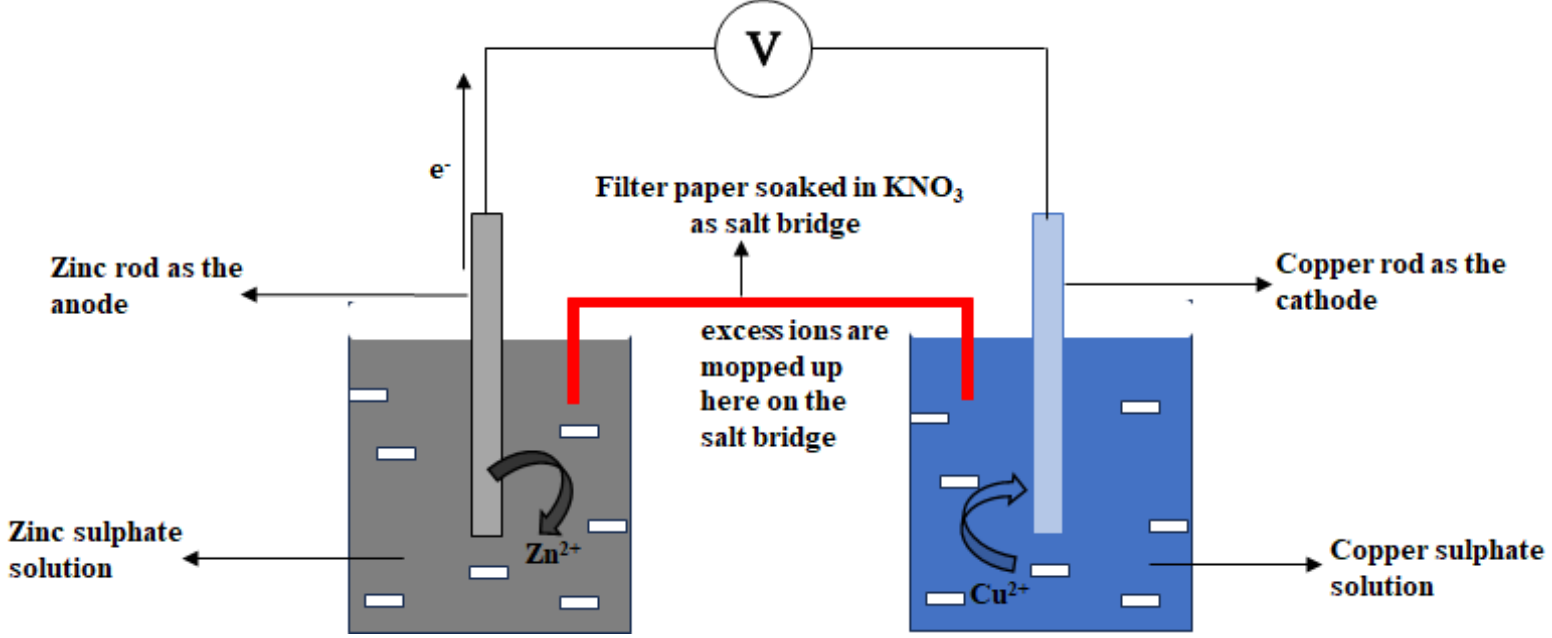


Fig. 3: Daniell cell

Figure 3 is the diagram of the Daniell cell. The chemical reaction in the cell is initiated from the anode where the more reducing zinc metal dissolves to give the Zn^{2+} ions according to the equation below:

$$Zn \rightarrow Zn^{2+} + 2e^{-}$$

The concentration of Zn^{2+} ions become high and so, excess Zn^{2+} ions are neutralized by NO_3^{-} ions that move out of the salt bridge or some Zn^{2+} ions may move into the salt bridge.

The copper ions in solution at the cathode pick up the electrons generated at the anode that flows through the connecting wire and become discharged or deposited according to this equation:

$Cu^{2+} + 2e^{-} \rightarrow Cu$

The concentration of Cu^{2+} ions become reduced and the excess SO_4^{2-} is neutralized by the K^{+} ions moving out of the salt bridge or the SO_4^{2-} may move into the salt bridge. The movement of electrons through the wire is called electric current. The current in Daniell or simple cell is a direct current and is high enough to light a bulb. The flow of current continues until the concentration of the Cu^{2+} ions, that is being reduced goes to zero. The net cell reaction is:

$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

At the standard conditions of 1 atm pressure, 25°C temperature and a unit activity, the Daniell cell can deliver about 1.1 volts. A short representation called the shorthand notation for the cell is:

$Zn/Zn^{2+}(m_1)//Cu^{2+}(m_2)/Cu$. The half-cell, Zn/Zn^{2+} is the anode, where oxidation occurs. Cu^{2+}/Cu is the cathode where reduction takes place. The single slash represents the interface between the solid metal and the solution, while the double slash depicts the salt bridge. The functions of the salt bridge are:

i. to complete the circuit by decongesting (through diffusion) excess ions from the half cells.

ii. to provide cations or anions to replace those consumed at the electrodes as well as use the ions to balance charges.

Standard half-cell potentials and reactions

A cell consists of two electrodes. The reduction potentials of the two electrodes determine the voltage of the cell. Each electrode possesses an intrinsic potential difference between the solid matrix and its salt in solution. Such reduction potential cannot be measured, yet the reduction potentials of single electrodes had been determined and are found tabulated in virtually all physical chemistry books. The feat looks like a paradox. It then becomes expedient to solve the paradox by choosing a standard that was given an arbitrary value. The standard electrode of convenience is the standard hydrogen electrode (SHE) which is arbitrarily assigned the value of standard electrode potential of zero at standard conditions. The half-cell representation and the reaction are $Pt/½H_2$ (1 atm)/H^+ (1 m) and $½H_2 \rightarrow H^+ + e^-$ respectively, with $E° = 0.00$ V.

The reduction potential of any electrode (half-cell) can be obtained by coupling such electrode with the SHE to form a cell. The voltage of the cell can be measured, and the value will be the reduction potential of the electrode coupled with the SHE. For example, the reduction potential of Cu^{2+}/Cu electrode is measured by setting up the cell, $Pt/½H_2(g)/H^+(aq)$ // $Cu^{2+}(aq)/Cu$. The half-cell reactions are:

$H_2 \rightarrow 2H^+ + 2e^-$ E° = 0.00 V (anode)

$Cu^{2+} + 2e- \rightarrow Cu$ E° = +0.34 V (cathode)

The full cell reaction is obtained by the addition of the half-cell reactions to give:

$H_2 + Cu^{2+} \rightarrow 2H^+ + Cu$ E^o_{cell} = +0.34 V

The value, E^o_{cell} = +0.34 V is the reduction potential of the single copper electrode since the anode or $½H_2$ (1 atm)/H^+ reduction potential is 0.00 V and that of the cathode (Cu^{2+}/Cu) is +0.34 V.

The shorthand notation for the Daniell cell is Zn/Zn^{2+} // Cu^{2+}/Cu or more generally for any Galvanic cell, A^{n+}/A // B^{x+}/B. If a cell notation is written such that the anode is on the left, one method of calculating the e.m.f. of such cells is, $E_{cell} = E_R - E_L$, where E_R and E_L are the reduction potentials of the right and left electrodes respectively. The other method is the use of half-cell or half-redox reaction equations. For example, for the Daniell cell, the half-cell reactions are,

$Zn \rightarrow Zn^{2+} + 2e^-$ $E° = 0.76$ V (anodic reaction)

$Cu^{2+} + 2e- \rightarrow Cu$ $E° = 0.34$ V (cathodic reaction)

Addition gives,

$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ $E^o_{\text{cell}} = 1.10$ V

Example:

Calculate the cell potentials whose cells are described by the following cell notations.

i. $\frac{1}{2}H_2/H^+$ // Cu^{2+}/Cu $\quad E^o_{H^+/\frac{1}{2}H_2}$ = 0.00 V, $E^o_{Cu^{2+}/Cu}$ = +0.34 V

ii. Ni^{2+}/Ni // $H^+/\frac{1}{2}H_2$ $\quad E^o_{Ni^{2+}/Ni}$ = +0.25 V, $E^o_{H^+/\frac{1}{2}H_2}$ = 0.00 V

iii. Zn/Zn^{2+} // Cu^{2+}/Cu $\quad E^o_{Zn^{2+}/Zn}$ = -0.76 V, $E^o_{Cu^{2+}/Cu}$ = +0.34 V

Solutions:

i. $E^o_{cell} = E^o_{red} - E^o_{oxi}$

Here, the reduction reaction is Cu^{2+}/Cu and the oxidation reaction is $\frac{1}{2}H_2/H^+$.

Therefore, $E^o_{cell} = 0.34 - 0 =$ +0.34 V.

ii. $E^o_{cell} = E^o_{red} - E^o_{oxi}$

Here, the reduction reaction is $H^+/\frac{1}{2}H_2$ and the oxidation reaction is Ni/Ni^{2+}.

Therefore, $E^o_{cell} = 0 - 0.25 =$ -0.25 V.

iii. $E^o_{\text{cell}} = E^o_{\text{red}} - E^o_{\text{oxi}}$

Here, the reduction reaction is Cu^{2+}/Cu and the oxidation reaction is Zn^{2+}/Zn.

Therefore, $E^o_{\text{cell}} = 0.34 - (-0.76) = +1.10$ V.

The physical significance of the electron generated in the arbitrary SHE ($½H_2/H^+$), are,

i.) its motion constitutes the source of electric current,

ii.) it acts as a reference electrode for the balancing of charges in redox reactions, and

iii.) it is the charged particle on which virtually all chemical principles and theories are predicted.

Concentration effects (Nernst equation):

The thermodynamics standard free energy change, ΔG°, is related to the standard e.m.f. of a cell by Eq. 19:

$$\Delta G^o = -nFE^o \qquad (19)$$

where *n* is the number of electrons involved in the cell reaction and *F* is the Faraday constant. If E_{cell} is positive, ΔG^o will be negative and the cell reaction will occur spontaneously and vice versa. For a reversible cell reaction at equilibrium, Eq. 20 is employed:

$$\Delta G^o = -RT \ln K \qquad (20)$$

where *K* is the equilibrium constant. By equating Eqs. 19 and 20 we have:

$$-nFE^o = -RT \ln K \Rightarrow E^o = \frac{RT}{nF} \ln K \qquad (21)$$

At the standard conditions of 1 atm, unit activity of electrolyte and 25°C, Eq. 21 becomes:

$$E^o = \frac{RT}{nF} \ln K = \frac{2.303RT}{nF} \log K = \frac{0.059}{n} \log K \qquad (22)$$

The quantity $\frac{2.303RT}{F}$ at 298 K is 0.059. Here R = 8.314 J/mol/K.

Example:

The reduction potentials of the electrodes of the cell, Fe/Fe^{2+} // Au^{3+}/Au are Fe^{2+}/Fe = -0.440 V and Au^{3+}/Au = 1.520 V,

i. Write the half-cell and the full cell reactions and state the polarity of the cell.

ii. Calculate the cell potential, the free energy change, the equilibrium constant and predict the feasibility of the cell reaction at 25°C.

Solution:

The oxidized electrode is the anode and it's recognized from the reduction potential which is usually less positive (more negative) than that of the reduced electrode which is the cathode.

$3Fe \rightarrow 3Fe^{2+} + 6e^-$ (anode/oxidation)

$2Au^{3+} + 6e^- \rightarrow 2Au$ (cathode/reduction)

Full cell reaction:

$3Fe + 2Au^{3+} \rightarrow 3Fe^{2+} + 2Au$

ii. $E_{\text{cell}} = E_{\text{red}} - E_{\text{oxi}} = 1.52 - (-0.44) = 1.52 + 0.44 = 1.960$ V

$\Delta G^o = -nFE^o = -6 \times 96500 \times 1.96 =$ -1134.8 kJmol^{-1}

$E^o = \frac{0.059}{n} \log K \Rightarrow \log K = \frac{6 \times 1.96}{0.059} = \frac{11.76}{0.059} = 199.3$

$\log K = 199.3 \qquad \Rightarrow K = 10^{199.3}$

E_{cell} is positive, ΔG^o is negative and K is high. Each of these parameters is an indicator that gives the direction of change, and for this example, each parameter shows the cell reaction to be spontaneous to the far right of the reaction.

If the electrolyte departs from the standard states of unity or the concentration of the electrolyte is less or greater than 1.0 mol.dm^{-3}, then the actual half-cell potentials must be calculated from the Nernst equation. The Nernst equation arose from experimental measurements, the results of which showed a linear relationship between the cell voltage and the electrolytes' concentrations. The Nernst equation is given by Eq. 23:

$$E = E^o - \frac{RT}{nF}\ln Q \qquad (23)$$

where Q is the reaction quotient of the half-cell reduction reaction.

For example, for the half-cell reduction of $Fe^{3+} + e^- \rightarrow Fe^{2+}$, $Q = \frac{Fe^{2+}}{Fe^{3+}}$. So,

$E_{\text{half-cell}} = E^o - \frac{RT}{nF}\ln\frac{[Fe^{2+}]}{[Fe^{3+}]} = E^o + \frac{RT}{nF}\ln\frac{[Fe^{3+}]}{[Fe^{2+}]}$ where E^o is the standard electrode potential if the electrode were to be operating under standard conditions.

At 298 K, $E_{\text{half-cell}} = E^o + \frac{0.059}{n} \log \frac{[\text{Fe}^{3+}]}{[\text{Fe}^{2+}]}$

Generally, $E_{\text{half-cell}} = E^o + \frac{RT}{nF} \ln \frac{[oxidixed\ form]}{[\text{reduced form}]}$

For the electrode, $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$, the voltage is,

$E_{\text{half-cell}} = E^o + \frac{RT}{nF} \ln \frac{[\text{MnO}_4^-][\text{H}^+]^8}{[\text{Mn}^{2+}]}$

The Nernst equation can also be used to calculate the cell voltage at the non-standard conditions. For example, the cell reaction, $Zn + Fe^{2+} \rightarrow Zn^{2+} + Fe$ generates E° = 0.32 V at 25°C (at standard condition). The Nernst equation for the cell (that is operating at no-standard condition) is,

$E_{\text{cell}} = 0.32 - \frac{0.059}{2} \log \frac{[\text{Zn}^{2+}]}{[\text{Fe}^{2+}]}$

Note that [Zn] and [Fe] are constants or are non-active species and are justifiably by the law of mass action (the law states that: the rate of reaction is proportional to the product of the concentration of reactants) omitted in the calculation of the equilibrium constant.

The reactant and product concentration (i.e. electrolyte and its product) do affect the cell voltage. If the reactant concentration is increased or the product concentration is decreased, the cell voltage will be more positive and vice versa. For example,

i.) Suppose $[Fe^{2+}] = 10^2$ mol dm^{-3} while $[Zn^{2+}]$ remains 1.0 mol dm^{-3}, then

$$E_{\text{cell}} = 0.32 - \frac{0.059}{2}\log\frac{1}{10^2} = 0.379 \text{ V}$$

ii.) $[Fe^{2+}]$ remains 1.0 mol dm^{-3} while $[Zn^{2+}]$ decreases to 0.10 mol dm^{-3}, then,

$$E_{\text{cell}} = 0.32 - \frac{0.059}{2}\log\frac{10^{-1}}{1} = 0.35 \text{ V}$$

The reverse of the above conditions are,

iii.) $[Fe^{2+}]$ decreases to 10^{-2} mol dm^{-3} while $[Zn^{2+}]$ remains the same as 1.0 mol dm^{-3},

$$E_{\text{cell}} = 0.32 - \frac{0.059}{2}\log 10^2 = 0.291 \text{ V}$$

iv.) $[Fe^{2+}]$ remains the same as 1.0 mol dm^{-3} while $[Zn^{2+}]$ increases to 10^3 mol dm^{-3},

$$E_{\text{cell}} = 0.32 - \frac{0.059}{2}\log 10^3 = 0.232 \text{ V}$$

v.) For $[Zn^{2+}] = [Fe^{2+}] = 1.0$ mol dm^{-3}, the cell voltage is,

$E_{\text{cell}} = 0.32 - \frac{0.059}{2}\log 1 = 0.320$ V (note that the concentration of these two ions fulfill the standard condition), hence the value 0.320 V obtained.

It can be seen that the driving force for the cell at $[Zn^{2+}] = [Fe^{2+}] = 1.0$ mol dm^{-3} is less than for cells (i) and (ii) but greater than for cells (iii) and (iv). The Nernst equation suggests that if the standard potential of a cell is zero, the concentration difference of the electrolytes of the two electrodes can still generate a voltage. This is the principle on which the functioning of the concentration cells is based. The cell, $Ag/Ag^{+}(0.001\ M) / Ag^{+}(0.100\ M)/Ag$ is a concentration cell. The anode is the more dilute electrode. The E^{o}_{cell} is zero because the standard reduction potentials, E° of both electrodes are equal. The half-cell reactions are:

$Ag \rightarrow Ag^+ + e^-$ (more dilute solution)

$Ag^+ + e^- \rightarrow Ag$ (more concentrated solution)

$Ag + Ag^+ (0.100\ M) \rightarrow Ag^+ (0.001\ M) + Ag$

$E^o_{cell} = Ag + Ag^+ (0.100\ M) \rightarrow Ag^+ (0.001\ M) + Ag = 0.00\ V$

$E_{cell} = 0.00 - \frac{0.059}{1}\log\frac{0.001}{0.100} = 0.118\ V$

The enthalpy change associated with the working of a cell can be obtained from the Gibbs-Helmholtz equation.

For an isothermal reaction, $\Delta G = \Delta H - T\Delta S$, a thermodynamic identity shows that $\Delta S = -\left(\frac{\partial \Delta G}{\partial T}\right)_P$. So, we can write $\Delta G = \Delta H + T\left(\frac{\partial \Delta G}{\partial T}\right)_P$. Recall also that $\Delta G = -n FE$, which gives $\left(\frac{\partial \Delta G}{\partial T}\right)_P = -nF\left(\frac{\partial E}{\partial T}\right)_P$. Therefore, $\Delta G = -nFE = \Delta H + T\left(\frac{\partial \Delta G}{\partial T}\right)_P = \Delta H - nFT\left(\frac{\partial E}{\partial T}\right)_P$ or $nFE = -\Delta H + nFT\left(\frac{\partial E}{\partial T}\right)_P$, giving $\Delta H = -nFE + nFT\left(\frac{\partial E}{\partial T}\right)_P = nF\left[T\left(\frac{\partial E}{\partial T}\right)_P - E\right]$. This last equation is known as the Gibbs-Helmholtz equation.

Example:

The reduction potential of the concentration cell $Ag/Ag^{+}(0.001\ M)\ /\ Ag^{+}(0.100\ M)/Ag$ is measured as a function of temperature as $E_T = 0.118 - 2.03 \times 10^{-4}(T - 295) - 4.5 \times 10^{-7}(T - 295)^2$. Calculate the heat generated by the chemical reaction occurring in the cell at 298K.

Solution:

At 298K, $E_{298} = 0.118 - 2.03 \times 10^{-4}(298 - 295) - 4.5 \times 10^{-7}(298 - 295)^2 = 0.118 - 2.03 \times 10^{-4}(3) - 4.5 \times 10^{-7}(3)^2 = 0.11738$ V.

$$E_T = 0.118 - 2.03 \times 10^{-4}T + 295 \times 2.03 \times 10^{-4} - 4.5 \times 10^{-7}[T(T - 295) - 295(T - 295)]$$

$$E_T = 0.118 - 2.03 \times 10^{-4}T + 295 \times 2.03 \times 10^{-4} - 4.5 \times 10^{-7}[T^2 - 590T - 87025]$$

$$E_T = 0.118 - 2.03 \times 10^{-4}T + 295 \times 2.03 \times 10^{-4} - 4.5 \times 10^{-7}T^2 + 4.5 \times 10^{-7} \times 590T + 4.5 \times 10^{-7} \times 87025$$

$$\left(\frac{\partial E}{\partial T}\right)_P = -2.03 \times 10^{-4} - 2 \times 4.5 \times 10^{-7}T + 4.5 \times 10^{-7} \times 590$$

$\left(\frac{\partial E}{\partial T}\right)_P$ at 298K,

$$\left(\frac{\partial E}{\partial T}\right)_P = -2.03 \times 10^{-4} - 2 \times 4.5 \times 10^{-7} \times (298) + 4.5 \times 10^{-7} \times 590 = -0.0002057\ VK^{-1}$$

$$\Delta H = nF\left[T\left(\frac{\partial E}{\partial T}\right)_P - E\right] = 1 \times 96500 \times [298 \times (-0.0002057) - 0.11738] = -17.24\ kJ\ mol^{-1}.$$